



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

2024/2025

SECP3843 – SPECIAL TOPIC IN DATA ENGINEERING

SECTION 02

ASSIGNMENT 1

LECTURER NAME: DR SEAN CHOON SEN

NAME	MATRIC ID
MUHAMMAD DANIEL HAKIM BIN SYAHRULNIZAM	A22EC0207

Table of Contents

1	Comparative Analysis of Data Professional Skills	3
2	Conclusion	4
3	References	5

Comparative Analysis of Data Professional Skills

Skill Category	Data Engineer	Data Analyst	Data Scientist
Core Function	Builds & maintains data infrastructure/pipelines	Interprets data for business insights	Develops models for predictions & strategic insights
Primary Languages	Python, Java, Scala, SQL	SQL, R or Python	Python, R, SQL (Java, Scala, C++ a bonus)
Data Management	SQL, NoSQL, Data Warehousing, ETL/ELT, Data Modeling, Streaming	SQL, Excel, Google Sheets (cleaning, basic handling)	SQL, Database Management, Data Wrangling
Big Data Tech	Hadoop, Spark, Hive, Kafka	Limited direct interaction	Hadoop, Spark, Hive
Cloud Platforms	AWS, Azure, Google Cloud (infrastructure)	Works with cloud analytics tools	Cloud computing tools (AWS, Azure, Google Cloud)
Analytics & Modeling	Data architecture, infrastructure design	Statistical analysis, basic predictive modeling, visualization	Machine Learning, Deep Learning, Statistics, Math, Data Mining
Key Tools	Airflow, dbt Cloud, Git	Tableau, Power BI, Excel, Jupyter	Tableau, Matplotlib, Jupyter, SPSS, SAS, Git
Crucial Soft Skills	Problem-solving, Communication, Collaboration, Business Acumen, Attention to Detail	Problem-solving, Communication, Business Acumen, Attention to Detail, Presentation	Problem-solving, Communication, Curiosity, Business Acumen, Data Ethics

Conclusion

As a summary, the main similarity among all three roles is the need for strong problem-solving skills and clear communication. All data professionals must understand business problems, work with data, and share their findings or solutions with different stakeholders. Being skilled in SQL and at least one programming language, either Python or R, is also a common requirement. These tools are essential for working with data at various stages of the data lifecycle (Edureka, 2022). In addition, having business insight and paying attention to detail are often mentioned as important soft skills. These skills stress the importance of understanding the business context and ensuring data accuracy (Northeastern University Graduate Programs, 2023).

However, the differences mainly lie in their main focus and level of technical knowledge. Data Engineers are the builders, concentrating on the infrastructure and flow of data. This requires deep knowledge of databases, distributed systems, and ETL processes (Simplilearn, 2024). Data Analysts are the interpreters who extract insights from existing data. They use statistical analysis and visualization tools to help with immediate business decisions (Coursera, 2022). Data Scientists are the innovators. They use advanced statistical modeling, machine learning, and programming to predict future trends and create complex data products (Simplilearn, 2024). While there may be overlaps, especially in smaller organizations where roles can be more generalized, these differences emphasize the unique contributions each role makes to the data ecosystem.

References

Coursera. (2022). *What does a data analyst do? Your 2025 career guide*. Retrieved from <https://www.coursera.org/articles/what-does-a-data-analyst-do-a-career-guide>

Edureka. (2022). *Difference between Data Analyst, Data Engineer and Data Scientist? Which among these is more difficult to become and which is a more interesting role?*.

Northeastern University Graduate Programs. (2023). *7 must-have skills for data analysts*. Retrieved from <https://graduate.northeastern.edu/knowledge-hub/data-analyst-skills/>

Simplilearn. (2024). *27 data science skills for a successful career in 2025*. Retrieved from <https://www.simplilearn.com/what-skills-do-i-need-to-become-a-data-scientist-article>

Simplilearn. (2024). *Data engineer skills, roles and responsibilities | 2025 guide*. Retrieved from <https://www.simplilearn.com/data-engineer-role-article>